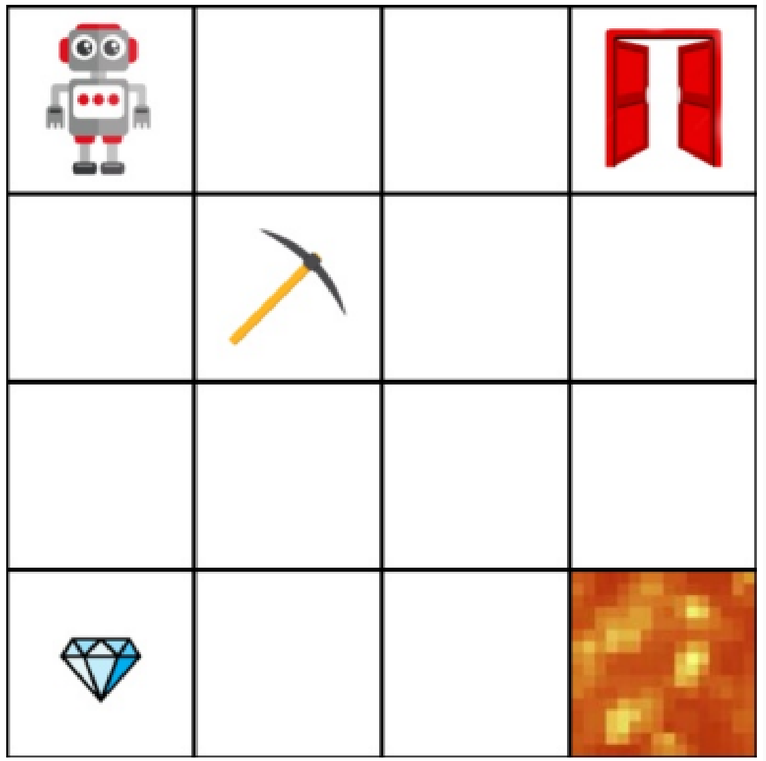
Neurosymbolic AI

Offline RL with Decision Transformers

Susanna Cifani

Environment and configurations

- 10 tasks defined via LTL
- States:
  - Image
  - Symbolic
- Rewards:
  - Three-Value Acceptance: Sparse rewards
  - Distance: Dense rewards
- Models: 3 Decision Transformers, each trained on a different dataset



Actions:

0 ↓

1 →

2 ↑

3 ←

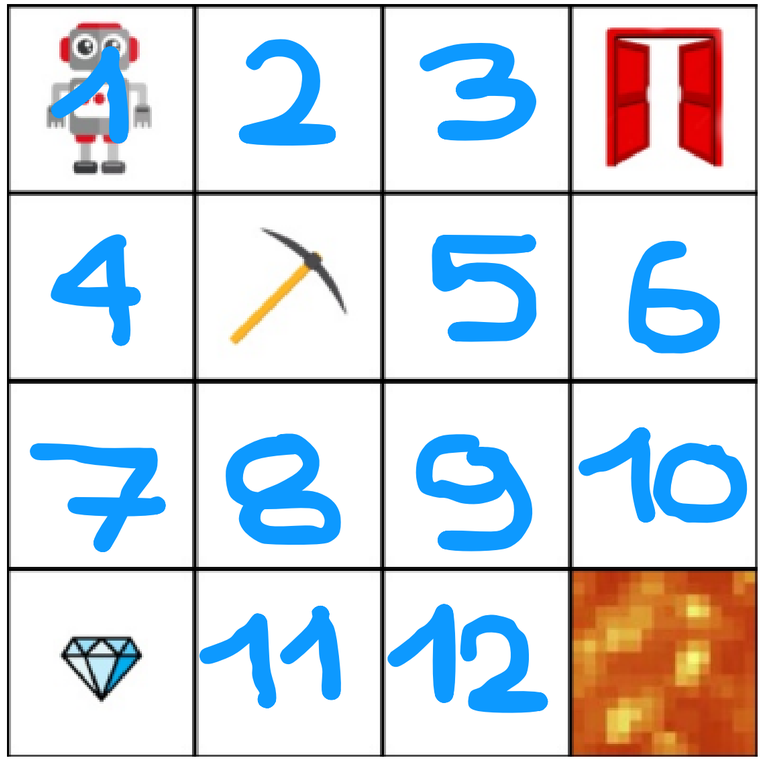
2 states X 2 rewards X 10 tasks X 3 DT

Dataset generation

- Generated ~2,000 trajectories per dataset

- Dijkstra's algorithm: Used to compute the shortest paths from each starting position
  - 12 empty starting cells x 84 paths extracted per cell

- Noise Injection (ε = 0.2): A 20% probability at each step to force the agent to take a random, sub-optimal action
  - *Why?* To teach the Decision Transformer how to recover from mistakes, rather than only memorizing perfect sequences



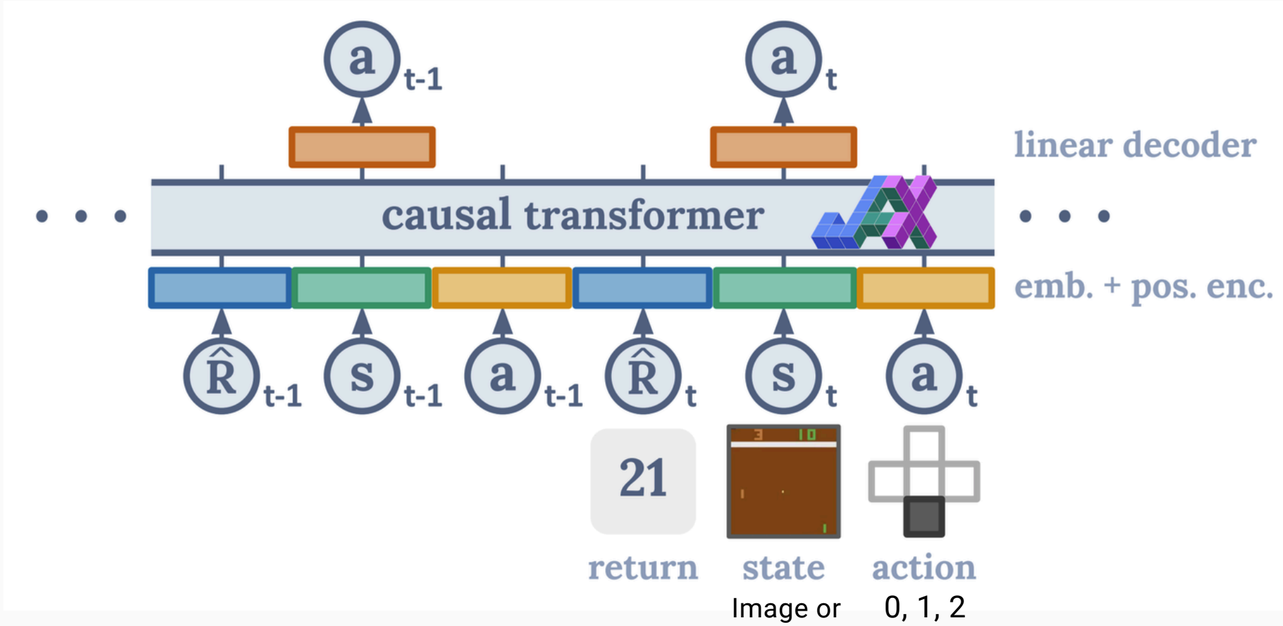
The 3 datasets

- Medium-Expert:
  - 50% Shortest trajectories
  - 50% Sub-optimal trajectories (20% noise applied to the shortest trajectories)
  - Outcome: 100% Successful trajectories
- Medium:
  - 100% Sub-optimal trajectories (20% noise)
  - Outcome: 100% Successful trajectories
- Medium-Replay:
  - 50% Sub-optimal trajectories (20% noise)
  - 50% Pure Random Walks
  - Outcome: 50% Successful trajectories, 50% Failed trajectories

Decision Transformers

- Context Window *K = 10*: The transformer looks at the last 10 timesteps to make a decision

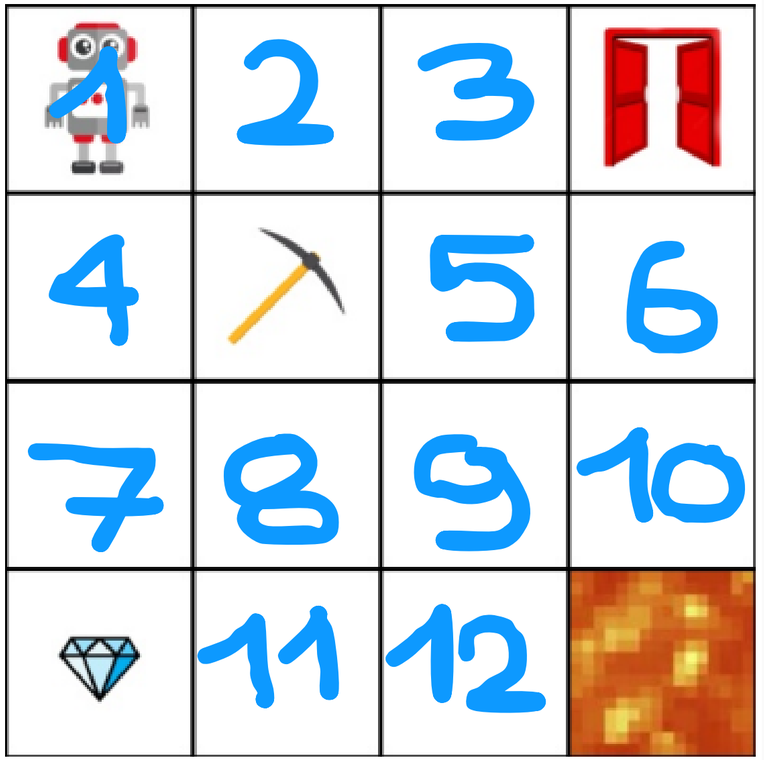
- Input:
  - **Return-To-Go (RTG)**: The target reward we want to achieve
  - **State**: The environment observation (Image/Symbolic)
  - **Action**: The move taken



Evaluation step

- Tested each possible starting point (empty cell) 10 times

- Tracked Metrics:
  - Average Reward (Raw)
  - Cumulative Discounted Reward (γ = 0.99)
  - Overall Success Rate (%)



Results

Symbolic state

- When provided with coordinates, the DT successfully learns LTL logic
- Near 100% Success Rate: Achieved across almost all 10 tasks, regardless of complexity
- High Robustness to Bad Data: Even the Medium-Replay dataset (containing 50% random failures) achieved near-perfect scores on most tasks

			three_value			distance		
			Reward	CDR	SR (%)	Reward	CDR	SR (%)
Symbolic	task1	dt_medium_expert	100.00 ± 0.00	92.28 ± 1.27	100.0%	100.00 ± 0.00	94.93 ± 0.70	100.0%
		dt_medium	83.33 ± 37.42	77.29 ± 34.73	83.3%	100.00 ± 0.00	94.93 ± 0.58	100.0%
		dt_medium_replay	100.00 ± 0.00	91.71 ± 3.15	100.0%	100.00 ± 0.00	95.87 ± 0.42	100.0%
	task2	dt_medium_expert	100.00 ± 0.00	91.66 ± 0.44	100.0%	100.00 ± 0.00	95.74 ± 0.51	100.0%
		dt_medium	100.00 ± 0.00	91.97 ± 0.69	100.0%	100.00 ± 0.00	95.51 ± 2.27	100.0%
		dt_medium_replay	75.00 ± 43.48	68.75 ± 39.86	75.0%	100.00 ± 0.00	96.21 ± 0.28	100.0%
	task3	dt_medium_expert	100.00 ± 0.00	91.05 ± 0.57	100.0%	100.00 ± 0.00	95.18 ± 0.39	100.0%
		dt_medium	100.00 ± 0.00	90.89 ± 0.46	100.0%	100.00 ± 0.00	95.03 ± 0.36	100.0%
		dt_medium_replay	100.00 ± 0.00	91.51 ± 0.63	100.0%	100.00 ± 0.00	95.03 ± 0.36	100.0%
	task4	dt_medium_expert	100.00 ± 0.00	89.61 ± 0.86	100.0%	100.00 ± 0.00	94.50 ± 0.69	100.0%
		dt_medium	100.00 ± 0.00	88.39 ± 2.84	100.0%	100.00 ± 0.00	94.38 ± 0.66	100.0%
		dt_medium_replay	100.00 ± 0.00	88.58 ± 1.41	100.0%	95.83 ± 9.36	90.23 ± 8.13	83.3%
	task5	dt_medium_expert	100.00 ± 0.00	91.05 ± 0.43	100.0%	100.00 ± 0.00	95.26 ± 0.59	100.0%
		dt_medium	100.00 ± 0.00	92.12 ± 0.92	100.0%	100.00 ± 0.00	95.42 ± 0.44	100.0%
		dt_medium_replay	58.33 ± 49.51	52.98 ± 44.97	58.3%	100.00 ± 0.00	94.85 ± 0.80	100.0%
	task6	dt_medium_expert	91.67 ± 27.75	82.91 ± 25.12	91.7%	100.00 ± 0.00	94.81 ± 0.85	100.0%
		dt_medium	100.00 ± 0.00	90.14 ± 1.07	100.0%	100.00 ± 0.00	94.65 ± 0.74	100.0%
		dt_medium_replay	75.00 ± 43.48	67.39 ± 39.10	75.0%	91.67 ± 11.83	86.35 ± 12.41	66.7%
	task7	dt_medium_expert	100.00 ± 0.00	91.66 ± 0.58	100.0%	100.00 ± 0.00	95.26 ± 0.61	100.0%
		dt_medium	100.00 ± 0.00	91.05 ± 0.57	100.0%	100.00 ± 0.00	95.03 ± 0.36	100.0%
		dt_medium_replay	100.00 ± 0.00	91.51 ± 0.63	100.0%	100.00 ± 0.00	95.41 ± 0.42	100.0%
	task8	dt_medium_expert	100.00 ± 0.00	90.89 ± 0.46	100.0%	100.00 ± 0.00	95.10 ± 0.48	100.0%
		dt_medium	100.00 ± 0.00	91.66 ± 0.58	100.0%	100.00 ± 0.00	94.95 ± 0.43	100.0%
		dt_medium_replay	100.00 ± 0.00	91.66 ± 0.58	100.0%	100.00 ± 0.00	95.03 ± 0.54	100.0%
	task9	dt_medium_expert	100.00 ± 0.00	90.89 ± 0.46	100.0%	100.00 ± 0.00	95.49 ± 0.46	100.0%
		dt_medium	100.00 ± 0.00	91.51 ± 0.63	100.0%	100.00 ± 0.00	95.11 ± 0.48	100.0%
		dt_medium_replay	100.00 ± 0.00	91.66 ± 0.58	100.0%	100.00 ± 0.00	95.18 ± 0.03	100.0%
	task10	dt_medium_expert	100.00 ± 0.00	91.05 ± 0.57	100.0%	100.00 ± 0.00	95.03 ± 0.54	100.0%
		dt_medium	100.00 ± 0.00	90.89 ± 0.46	100.0%	100.00 ± 0.00	95.15 ± 0.50	100.0%
		dt_medium_replay	100.00 ± 0.00	91.05 ± 0.57	100.0%	100.00 ± 0.00	95.41 ± 0.42	100.0%

- task1: visit({0}, {1})
- task2: visit({0}, {1}, {2})
- task3: seq_visit({0}, {1})
- task4: seq_visit({0}, {1}) + seq_visit({2}, {3})
- task5: seq_visit({0}, {1}) + visit({2})
- task6: seq_visit({0}, {1}) + visit({2}, {3})
- task7: visit({0}, {1}) + glob_av({2})
- task8: visit({0}, {1}) + glob_av({2}) + glob_av({3})
- task9: seq_visit({0}, {1}) + glob_av({2})
- task10: seq_visit({0}, {1}) + glob_av({2}) + glob_av({3})

Image state (three-value acceptance)

Visit (task 1, 2)

- Visit items in a random order
- Behavior: High performance, vulnerable to noisy datasets

Sequential Visit (task 3, 4, 5, 6, 9, 10)

- Visit items in a sequential order
- Behavior: Low performance on pure sequences (3, 4) and sequences with avoidance (9, 10). Adding flexible visits (5, 6) helps Expert performance

Global Avoidance (task 7, 8, 9, 10)

- Visit items, while avoiding others
- Behavior: Kills performance, Medium tends to perform the best

			three_value			distance		
			Reward	CDR	SR (%)	Reward	CDR	SR (%)
State	Task	Model						
Image	task1	dt_medium_expert	100.00 ± 0.00	86.91 ± 5.53	100.0%	100.00 ± 0.00	94.87 ± 2.33	100.0%
		dt_medium	90.00 ± 30.13	76.77 ± 26.66	90.0%	92.92 ± 17.51	86.67 ± 15.22	85.8%
		dt_medium_replay	45.83 ± 50.04	38.02 ± 41.63	45.8%	28.33 ± 33.51	25.53 ± 29.55	10.0%
	task2	dt_medium_expert	100.00 ± 0.00	89.63 ± 3.14	100.0%	100.00 ± 0.00	94.27 ± 1.24	100.0%
		dt_medium	100.00 ± 0.00	86.14 ± 5.18	100.0%	74.17 ± 13.98	70.08 ± 12.01	22.5%
		dt_medium_replay	40.83 ± 49.36	31.17 ± 37.82	40.8%	28.89 ± 43.20	26.83 ± 39.84	25.8%
	task3	dt_medium_expert	0.00 ± 0.00	0.00 ± 0.00	0.0%	13.75 ± 22.42	13.06 ± 21.47	0.0%
		dt_medium	0.00 ± 0.00	0.00 ± 0.00	0.0%	45.83 ± 13.88	40.86 ± 12.49	0.0%
		dt_medium_replay	0.00 ± 0.00	0.00 ± 0.00	0.0%	25.83 ± 25.09	22.91 ± 22.37	0.0%
	task4	dt_medium_expert	38.33 ± 48.82	28.48 ± 36.59	38.3%	30.00 ± 15.06	27.39 ± 14.71	0.0%
		dt_medium	7.50 ± 26.45	6.42 ± 22.65	7.5%	21.46 ± 34.50	17.94 ± 29.13	11.7%
		dt_medium_replay	2.50 ± 15.68	2.12 ± 13.31	2.5%	9.17 ± 24.67	8.04 ± 21.56	0.8%
	task5	dt_medium_expert	100.00 ± 0.00	84.04 ± 3.25	100.0%	73.33 ± 44.41	62.22 ± 38.20	73.3%
		dt_medium	69.17 ± 46.37	54.20 ± 37.10	69.2%	40.56 ± 34.35	38.10 ± 32.29	21.7%
		dt_medium_replay	35.00 ± 47.90	27.70 ± 37.95	35.0%	19.17 ± 30.76	18.01 ± 27.82	10.0%
	task6	dt_medium_expert	96.67 ± 18.03	78.58 ± 16.72	96.7%	82.08 ± 28.10	71.54 ± 23.90	64.2%
		dt_medium	100.00 ± 0.00	81.01 ± 4.35	100.0%	45.42 ± 14.85	41.08 ± 12.77	3.3%
		dt_medium_replay	4.17 ± 20.07	2.93 ± 14.21	4.2%	31.67 ± 32.04	29.41 ± 28.44	10.0%
	task7	dt_medium_expert	3.33 ± 31.58	2.90 ± 29.61	6.7%	46.67 ± 88.81	43.37 ± 85.26	73.3%
		dt_medium	58.33 ± 81.56	49.72 ± 76.73	79.2%	45.00 ± 89.68	42.83 ± 85.30	72.5%
		dt_medium_replay	42.50 ± 49.64	39.34 ± 45.96	42.5%	41.67 ± 40.13	39.22 ± 37.40	25.0%
	task8	dt_medium_expert	0.83 ± 9.13	0.80 ± 8.77	0.8%	100.00 ± 0.00	93.50 ± 2.57	100.0%
		dt_medium	25.00 ± 89.11	20.49 ± 83.40	55.0%	57.08 ± 74.84	57.23 ± 65.39	66.7%
		dt_medium_replay	17.50 ± 38.16	15.30 ± 33.66	17.5%	59.17 ± 50.62	54.61 ± 46.76	56.7%
	task9	dt_medium_expert	0.00 ± 0.00	0.00 ± 0.00	0.0%	50.00 ± 0.00	46.90 ± 2.67	0.0%
		dt_medium	0.00 ± 0.00	0.00 ± 0.00	0.0%	20.00 ± 60.25	19.27 ± 57.93	0.0%
		dt_medium_replay	1.67 ± 12.86	1.29 ± 9.98	1.7%	50.00 ± 0.00	48.46 ± 2.08	0.0%
	task10	dt_medium_expert	0.00 ± 0.00	0.00 ± 0.00	0.0%	50.00 ± 0.00	48.70 ± 0.99	0.0%
		dt_medium	10.00 ± 83.41	5.48 ± 76.74	40.0%	46.25 ± 23.52	45.65 ± 16.50	0.0%
		dt_medium_replay	0.00 ± 0.00	0.00 ± 0.00	0.0%	37.50 ± 21.74	35.57 ± 20.73	0.0%

Image state (distance)

- task1: visit({0}, {1})
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- task10: seq_visit({0}, {1}) + glob_av({2}) + glob_av({3})

Visit (task 1, 2)

- Performance similar to three-value acceptance for Medium-Expert, but worse on noisy datasets

Sequential Visit (task 3, 4, 5, 6, 9, 10)

- Worse performance than three-value acceptance

Global Avoidance (task 7, 8, 9, 10)

- Better performance than three-value acceptance

			three_value			distance		
			Reward	CDR	SR (%)	Reward	CDR	SR (%)
State Image	Task	Model						
	task1	dt_medium_expert	100.00 ± 0.00	86.91 ± 5.53	100.0%	100.00 ± 0.00	94.87 ± 2.33	100.0%
		dt_medium	90.00 ± 30.13	76.77 ± 26.66	90.0%	92.92 ± 17.51	86.67 ± 15.22	85.8%
		dt_medium_replay	45.83 ± 50.04	38.02 ± 41.63	45.8%	28.33 ± 33.51	25.53 ± 29.55	10.0%
	task2	dt_medium_expert	100.00 ± 0.00	89.63 ± 3.14	100.0%	100.00 ± 0.00	94.27 ± 1.24	100.0%
		dt_medium	100.00 ± 0.00	86.14 ± 5.18	100.0%	74.17 ± 13.98	70.08 ± 12.01	22.5%
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		dt_medium	0.00 ± 0.00	0.00 ± 0.00	0.0%	45.83 ± 13.88	40.86 ± 12.49	0.0%
		dt_medium_replay	0.00 ± 0.00	0.00 ± 0.00	0.0%	25.83 ± 25.09	22.91 ± 22.37	0.0%
	task4	dt_medium_expert	38.33 ± 48.82	28.48 ± 36.59	38.3%	30.00 ± 15.06	27.39 ± 14.71	0.0%
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	task8	dt_medium_expert	0.83 ± 9.13	0.80 ± 8.77	0.8%	100.00 ± 0.00	93.50 ± 2.57	100.0%
		dt_medium	25.00 ± 89.11	20.49 ± 83.40	55.0%	57.08 ± 74.84	57.23 ± 65.39	66.7%
		dt_medium_replay	17.50 ± 38.16	15.30 ± 33.66	17.5%	59.17 ± 50.62	54.61 ± 46.76	56.7%
	task9	dt_medium_expert	0.00 ± 0.00	0.00 ± 0.00	0.0%	50.00 ± 0.00	46.90 ± 2.67	0.0%
		dt_medium	0.00 ± 0.00	0.00 ± 0.00	0.0%	20.00 ± 60.25	19.27 ± 57.93	0.0%
		dt_medium_replay	1.67 ± 12.86	1.29 ± 9.98	1.7%	50.00 ± 0.00	48.46 ± 2.08	0.0%
	task10	dt_medium_expert	0.00 ± 0.00	0.00 ± 0.00	0.0%	50.00 ± 0.00	48.70 ± 0.99	0.0%
		dt_medium	10.00 ± 83.41	5.48 ± 76.74	40.0%	46.25 ± 23.52	45.65 ± 16.50	0.0%
		dt_medium_replay	0.00 ± 0.00	0.00 ± 0.00	0.0%	37.50 ± 21.74	35.57 ± 20.73	0.0%

Thanks